

Variational Neurons and Bayesian Neural Networks: Distinct Probabilistic Objects, Inference Loci, and Computational Granularity

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Abstract. Probabilistic neural architectures frequently reuse the same mathematical devices—priors, Kullback–Leibler (KL) terms, reparameterized samples, Monte Carlo evaluation, and predictive distributions—while assigning them to different random objects and computational loci. This article provides a clarifying taxonomy for these constructions and formalizes an operational notion of the variational neuron. Four non-equivalent axes are separated: Bayesianity, variationality, stochasticity, and distributional computation. Six reporting questions then identify the random object, inferential or regularization locus, conditioning information, sampling location, propagated representation, and prediction rule. A standard Bayesian neural network (BNN) represents dataset-conditioned uncertainty over parameters or functions and predicts by posterior marginalization. By contrast, the Elemental Variational Expanse (EVE) denotes a unit-level regime in which an input-conditioned latent distribution is placed inside a declared hidden computational unit, contributes a separately measurable variational term, participates explicitly in the forward map, and exposes unit-indexed diagnostics. The framework distinguishes a strict variational latent neuron, whose distribution approximates a posterior under an explicit latent-variable model, from a variationally regularized distributional neuron trained end-to-end without an independent local likelihood. It also positions variational neurons relative to major neighboring probabilistic architectures. A reproducible five-seed heteroscedastic benchmark serves as a controlled illustration: EVE attains lower mean negative log-likelihood (NLL) and continuous ranked probability score (CRPS) than a capacity-matched deterministic cell at similar mean squared error (MSE), while its narrower intervals accompany slightly lower coverage. These results demonstrate distinct probabilistic utility arising from distributional computation at the hidden-unit level.

Keywords: Bayesian neural networks · variational inference · variational neurons · probabilistic computation · computational granularity · uncertainty quantification · EVE

1 Introduction

Neural networks can be made probabilistic in several fundamentally different ways. A model may place a distribution over network parameters, introduce latent variables at the data, layer, sequence, or unit level, randomize hidden activations, predict an output distribution, average independently trained networks, or combine several of these mechanisms. These constructions can all produce samples or uncertainty scores, yet they need not describe the same unknown quantity or perform the same form of inference. The terms *Bayesian*, *variational*, *stochastic*, *probabilistic*, and *distributional* should therefore not be treated as interchangeable [1,17].

Shared devices—priors, KL terms, reparameterization, and Monte Carlo sampling—do not imply shared probabilistic semantics. A standard BNN represents dataset-conditioned uncertainty over parameters or functions and predicts by marginalizing those unknowns; a variational neuron instead places an input-conditioned local latent distribution inside hidden computation. The decisive distinction is the probabilistic object and its locus.

This distinction does not make the two constructions mutually exclusive. A local variational state may be embedded in a broader Bayesian latent-variable model, and a network may combine uncertainty over weights with local latent variability. The correct label follows from the explicitly declared random objects, conditioning evidence, inferential targets, and prediction rule.

This article makes three contributions: (i) a four-axis taxonomy organized by six reporting questions; (ii) an operational definition of EVE as an addressable hidden-unit variational primitive, distinguishing posterior inference from local variational regularization; and (iii) a controlled comparison with deterministic and output-probabilistic alternatives.

The formal contribution is an operational criterion for attributing variational computation to an individually addressable hidden unit, distinct from layer-, representation-, parameter-, and output-level formulations.

Table 1. Established foundations and the contribution formalized in this article.

Element	Established foundation	Contribution formalized here
Probabilistic taxonomy	Bayesian deep-learning and uncertainty-quantification surveys	Four non-equivalent axes combined with six operational questions that identify the random object and its computational path
Local latent computation	variational autoencoders (VAEs), deep latent-variable models, stochastic hidden units, and information bottlenecks	An addressable unit abstraction with declared conditioning, local reference distribution, decomposable variational cost, explicit forward role, and unit-indexed diagnostics
Controlled illustration	Deterministic, heteroscedastic, and stochastic regression benchmarks	A capacity-matched example showing how hidden local distributional computation and output-level probabilistic prediction occupy different loci

2 Four Distinct Axes of Probabilistic Neural Computation

The taxonomy is multi-label: each architecture is evaluated independently along the four axes before any global label is assigned. Section 5 applies this framework to representative probabilistic architectures.

Each construction is described through six fields:

1. Which object is random?
2. Where is inference or regularization located?
3. On which evidence is the distribution conditioned?
4. Where does sampling occur?
5. What representation is propagated to the next computation?
6. How is the final prediction formed?

A candidate axis is retained only when architectures can differ on that property without a necessary corresponding change on the others. Bayesian updating, variational optimization, sampled computation, and distribution-valued representation therefore remain distinct.

2.1 Bayesianity

Bayesian modelling specifies uncertain quantities, assigns priors, conditions on observations, and forms posterior or posterior-predictive quantities by integration. In neural networks, the uncertain quantities may be weights, functions, hyperparameters, local latent variables, or combinations thereof [5,29,33,48]. Bayesianity is therefore a property of the probabilistic model and its inferential semantics, not of one particular neural architecture.

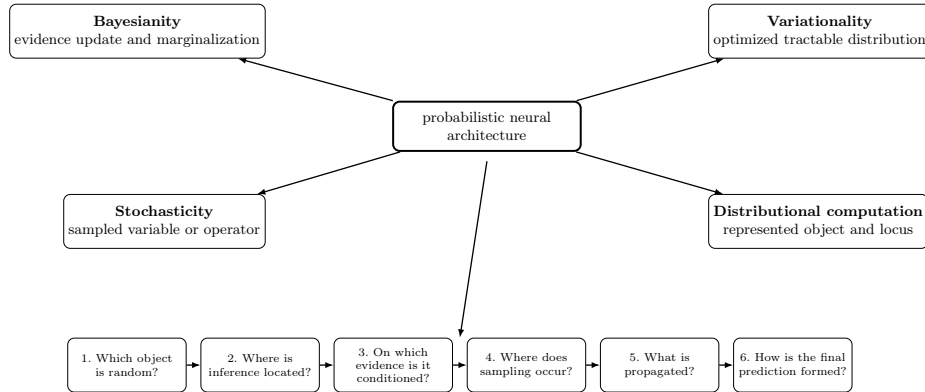


Fig. 1. Classification map for probabilistic neural architectures. The four axes are non-equivalent descriptive dimensions. The six questions identify the random object, inferential locus, conditioning evidence, sampling location, propagated representation, and prediction rule.

2.2 Variationality

Variational inference (VI) replaces an intractable inference problem by optimization over a tractable family [10]. It can approximate a posterior over weights, a posterior over latent variables, a filtering distribution, or another target density. Conversely, Bayesian inference can be performed by Markov chain Monte Carlo (MCMC), Laplace approximation, expectation propagation, sequential Monte Carlo, or other methods without VI [19,24,40,46]. Variationality is therefore not synonymous with Bayesianity.

2.3 Stochasticity

A stochastic computation samples a random variable or uses a randomized operator. Dropout masks, noisy activations, latent codes, randomized gates, and posterior weight samples all satisfy this broad criterion. Stochasticity alone does not identify the source, inferential meaning, or uncertainty semantics of the randomness [7,23,30,47].

2.4 Distributional computation and granularity

A deterministic neuron maps an input representation to a point-valued activation. A distributional neuron parameterizes or samples from a local distribution that participates in the forward computation. This architectural property does not by itself determine whether the distribution represents epistemic uncertainty, aleatoric uncertainty, ambiguity, controlled variability, compression, or another learned latent structure.

Table 2. Four complementary concepts in probabilistic neural computation.

Concept	Defining question	Typical mathematical object	Properties requiring separate identification
Bayesianity	Which unknowns receive priors, are conditioned on data, and are marginalized?	$p(\theta \mid \mathcal{D})$, $p(z \mid x, \mathcal{D})$, posterior-predictive integrals	Approximation algorithm, distributional family, and neural implementation
Variationality	Which tractable distribution is optimized relative to a target density?	$q_\lambda(\theta)$ or $q_\phi(z \mid x)$ and an evidence lower bound (ELBO)	Bayesian semantics and interpretation of the target density
Stochasticity	Which random variable or randomized operator is sampled?	Noise ϵ , masks, gates, weights, or latent states	Posterior semantics, calibration, and uncertainty interpretation
Distributional computation	At what architectural location is a distribution represented and used?	A local family $q_i(z_i \mid h_i)$, a sample, or distributional statistics	Random object, conditioning, propagated representation, and prediction rule

The declared random object, sampling site, propagated representation, and decision output are distinct descriptive roles:

$$\text{random object} \neq \text{sampling location} \neq \text{propagated object} \neq \text{decision output}. \quad (1)$$

For example, the local reparameterization trick samples a pre-activation even though the modeled random object remains a weight posterior. Conversely, a local latent distribution may be sampled inside a unit and reduced to a point-valued activation before the next layer. Sampling location therefore does not determine inferential identity.

The non-implications are immediate. A model can be Bayesian but not variational, as in an MCMC-trained BNN; variational but not a BNN, as in a VAE with point-estimated weights; stochastic but neither Bayesian nor variational, as in fixed noise injection; or distributional only at the output, as in heteroscedastic regression. Hybrid architectures may combine all four properties.

3 Bayesian Neural Networks: Posterior Uncertainty over Models

3.1 Canonical formulation

Let $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ be a dataset and let w denote network parameters. A canonical BNN specifies a prior $p(w)$ and likelihood $p(\mathcal{D} \mid w) = \prod_n p(y_n \mid x_n, w)$. Bayes' rule gives

$$p(w \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid w)p(w)}{p(\mathcal{D})}, \quad p(\mathcal{D}) = \int p(\mathcal{D} \mid w)p(w) dw. \quad (2)$$

For a new input x_* ,

$$p(y_* \mid x_*, \mathcal{D}) = \int p(y_* \mid x_*, w) p(w \mid \mathcal{D}) dw. \quad (3)$$

The defining operation is posterior marginalization over plausible models or functions, not merely the existence of a distribution somewhere in the network [33,48].

Exact inference is generally intractable. Bayes by Backprop learns a variational distribution over weights [11,18]; probabilistic backpropagation uses assumed-density and expectation-propagation ideas [19]; stochastic gradient Langevin dynamics (SGLD) approximates posterior sampling [46]; Laplace methods approximate the posterior around a mode [40]; and Stochastic Weight Averaging-Gaussian (SWAG) fits a Gaussian approximation from an optimization trajectory [31]. These methods differ, but their inferential object remains model parameters or induced functions conditioned on the dataset.

3.2 Weight space, function space, and uncertainty

Different weight configurations can encode the same or nearly the same function, and simple weight-space priors can induce complex function-space priors [5,15]. Gaussian processes provide a direct function-space construction and reveal limiting relationships with infinitely wide networks [33,37]. Operationally, BNN uncertainty concerns posterior uncertainty over predictions or functions after conditioning on data.

In heteroscedastic regression,

$$y \mid x, w \sim \mathcal{N}(\mu_w(x), \sigma_w^2(x)), \quad (4)$$

uncertainty in w and the conditional variance $\sigma_w^2(x)$ play different roles. The law of total variance gives

$$\text{Var}(y_* \mid x_*, \mathcal{D}) = \mathbb{E}_{p(w|\mathcal{D})}[\text{Var}(y_* \mid x_*, w)] \quad (5)$$

$$+ \text{Var}_{p(w|\mathcal{D})}(\mathbb{E}[y_* \mid x_*, w]). \quad (6)$$

The first term is commonly associated with expected data uncertainty and the second with model uncertainty, subject to model adequacy and identifiability [14,20,25]. Calibration and robustness remain empirical properties; approximate BNNs can fail under misspecification, and deep ensembles provide an important non-Bayesian reference [21,28,35].

4 Variational Neurons: Local Distributional Computation

4.1 Variational inference and amortized latent variables

For observed variables x and unknowns u with joint density $p(x, u)$, VI selects a tractable family $\mathcal{Q}$ and solves

$$q^*(u) = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(u) \parallel p(u \mid x)). \quad (7)$$

Equivalently, it maximizes

$$\mathcal{L}(q) = \mathbb{E}_{q(u)}[\log p(x, u)] - \mathbb{E}_{q(u)}[\log q(u)] = \log p(x) - \text{KL}(q(u) \parallel p(u \mid x)). \quad (8)$$

The optimization machinery is agnostic to whether u denotes weights, a global code, a trajectory, or a local hidden state [9,10].

A VAE specifies $p_\theta(x, z) = p(z)p_\theta(x | z)$ and an amortized family $q_\phi(z | x)$ [27,39]:

$$\mathcal{L}_{\text{VAE}}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x | z)] - \text{KL}(q_\phi(z | x) \| p(z)). \quad (9)$$

For a diagonal Gaussian,

$$q_\phi(z | x) = \mathcal{N}(\mu_\phi(x), \text{Diag}(\sigma_\phi^2(x))), \quad z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (10)$$

Importance weighting and normalizing flows provide richer variational families [12,38]. A conventional VAE usually places the latent at the example or model-component level. The variational neuron applies analogous machinery at a declared hidden computational unit.

A KL term alone does not establish Bayesian posterior inference. For

$$\mathcal{J} = \mathcal{L}_{\text{task}} + \beta \text{KL}(q_\phi(z | h) \| p(z)), \quad (11)$$

the expression is a negative ELBO when the task term is the negative expected log-likelihood of a specified joint model, $\beta = 1$, and q_ϕ approximates the corresponding posterior. Otherwise, it is more accurately described as variational regularization.

4.2 Local latent computation

Let $h_i \in \mathbb{R}^{d_i}$ be the deterministic input to hidden unit i . A minimal EVE unit introduces $z_i \in \mathbb{R}^{k_i}$ and

$$q_{\phi_i}(z_i | h_i, c_i) = \mathcal{N}(\mu_{\phi_i}(h_i, c_i), \text{Diag}(\sigma_{\phi_i}^2(h_i, c_i))), \quad (12)$$

where c_i may contain contextual, layer-level, or temporal information. A sample is obtained by

$$z_i = \mu_{\phi_i}(h_i, c_i) + \sigma_{\phi_i}(h_i, c_i) \odot \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I), \quad (13)$$

and participates explicitly in

$$a_i = F_{\eta_i}(h_i, z_i). \quad (14)$$

A prior or reference distribution $p_{\psi_i}(z_i | c_i)$ constrains the local state. It may be standard normal, contextual, hierarchical, or temporal [13,36,44].

For a set $\mathcal{I}$ of declared units, a common end-to-end objective is

$$\mathcal{J}_{\text{EVE}} = \mathbb{E}_\epsilon[\ell_{\text{task}}(f(x; z), y)] + \sum_{i \in \mathcal{I}} \beta_i \text{KL}(q_{\phi_i}(z_i | h_i, c_i) \| p_{\psi_i}(z_i | c_i)) + \Omega, \quad (15)$$

where $\beta_i \geq 0$ weights the local KL contribution of unit i , and Ω may contain local capacity, homeostatic, temporal, or anti-collapse constraints. For a conditionally

factorized layer,

$$q_{\Phi}(\mathbf{z} \mid \mathbf{h}, \mathbf{c}) = \prod_{i \in \mathcal{I}} q_{\phi_i}(z_i \mid h_i, c_i),$$

$$\mathcal{K}_{\text{layer}} = \sum_{i \in \mathcal{I}} \text{KL}(q_{\phi_i}(z_i \mid h_i, c_i) \parallel p_{\psi_i}(z_i \mid c_i)). \quad (16)$$

This factorization represents multiple local variational processes. Units usually output a point activation from a sample or declared statistic; propagation of distribution parameters must be declared explicitly.

4.3 Operational computational granularity

Neuron-level granularity is anchored to a declared partition $\mathcal{P} = \{I_1, \dots, I_J\}$ of local latent coordinates.

Definition 1 (Addressable variational unit). *A latent component indexed by i is an addressable variational unit when: (i) the model contains an explicitly indexed local variable z_i and conditional family $q_{\phi_i}(z_i \mid h_i, c_i)$; (ii) a prior or reference distribution is declared for that variable; (iii) its inferential or regularization contribution is separately computable, even when parameters are shared; (iv) the forward map depends explicitly on z_i ; (v) diagnostics such as local scale, divergence, mean energy, or utilization can be reported at the same index; and (vi) the partition, latent dimension, and parameter-sharing convention are part of the model specification.*

A k -dimensional latent vector is one unit when its distribution, reference, cost, forward role, and diagnostics are jointly defined. It represents k scalar units only when coordinate-wise or block-wise factorization is explicitly declared and the corresponding quantities are separately reportable. An inseparable joint latent with only aggregate cost and diagnostics is therefore a block- or layer-level latent rather than a collection of neurons.

Definition 2 (Operational EVE signature). *A construction satisfies the operational EVE signature if and only if every declared unit satisfies Definition 1 under the stated partition.*

Here, *neuron* refers to the declared computational granularity of the model, independently of its software implementation.

Consequences of addressability. Under Eq. (16), let $\mathcal{K}_i = \text{KL}(q_{\phi_i}(z_i \mid h_i, c_i) \parallel p_{\psi_i}(z_i \mid c_i))$, with $\mathcal{K}_{\text{layer}} = \sum_{i \in \mathcal{I}} \mathcal{K}_i$. The shared index i aligns latent state, local cost, forward dependence, and diagnostics, enabling unit-wise decomposition even with parameter sharing. Equation (14) also defines three local policies: $z_i \sim q_{\phi_i}$, $z_i = \mu_{\phi_i}$, and $z_i \sim p_{\psi_i}$, corresponding to learned-state sampling, mean-state execution, and reference-state intervention.

Algorithm 1 makes explicit the unit computation represented by Eqs. (12)–(15).

Algorithm 1 Addressable EVE unit: forward computation and local contribution.

Require: h_i, c_i

- 1: Obtain μ_i, σ_i from $q_{\phi_i}(z_i | h_i, c_i)$
 - 2: $\epsilon_i \sim \mathcal{N}(0, I)$; $z_i \leftarrow \mu_i + \sigma_i \odot \epsilon_i$
 - 3: $a_i \leftarrow F_{\eta_i}(h_i, z_i)$
 - 4: $\mathcal{K}_i \leftarrow \text{KL}(q_{\phi_i}(z_i | h_i, c_i) \| p_{\psi_i}(z_i | c_i))$
 - 5: **return** $(a_i, \mathcal{K}_i)$
-

4.4 Strict posterior inference and local variational regularization

Two regimes must be distinguished.

Strict probabilistic reading. There is an explicit observed or declared local target a_i^* , with a local likelihood or factor $p(a_i^* | z_i, h_i, c_i)$, and a joint model under which $q_{\phi_i}(z_i | h_i, c_i, a_i^*)$ approximates a posterior. For unit coefficient one,

$$\mathcal{L}_i = \mathbb{E}_{q_{\phi_i}}[\log p_{\eta_i}(a_i^* | z_i, h_i, c_i)] - \text{KL}(q_{\phi_i} \| p_{\psi_i}). \quad (17)$$

This is a genuine local ELBO. A coefficient $\beta_i \neq 1$ gives a weighted variational objective.

End-to-end architectural reading. The unit has no independent target a_i^* . Its local state is composed by later layers and trained through the downstream task likelihood. The data-fit term is global while the KL contribution is local. Equation (15) is then described as *locally factorized variational regularization inside a globally trained stochastic network*. The term *posterior* should be reserved for a declared posterior target.

Accordingly, a *variational latent neuron* approximates a posterior under an explicit latent-variable model, whereas an addressable EVE unit operating under the end-to-end architectural reading is termed a *variationally regularized distributional neuron* (EVE-reg): it uses an input-conditioned local distribution and local reference penalty without claiming posterior semantics.

4.5 Deterministic policy, dimensionality, and observability

At the forward-computation level, $\sigma_{\phi_i}(h_i, c_i) \rightarrow 0$ concentrates the local distribution toward a point state. This is not generally a finite-objective limit under a fixed non-degenerate prior because $\text{KL}(\mathcal{N}(\mu, \sigma^2) \| \mathcal{N}(0, I))$ diverges as $\sigma \rightarrow 0$. Separately, replacing the sampled state by its mean defines a deterministic deployment policy even when the learned distribution remains non-degenerate. Functional concentration and mean-state deployment are therefore different cases.

The latent dimension k_i is not network width. Width counts units; k_i counts local latent degrees of freedom. The atomic case $k_i = 1$ defines one stochastic coordinate, whereas $k_i > 1$ defines a local latent subspace.

Suppressing contextual arguments for readability, let $\Sigma_i = \text{Diag}(\sigma_i^2)$ denote the local covariance. Useful diagnostics include

$$\mathcal{K}_i, \quad \|\mu_i\|^2, \quad \text{tr } \Sigma_i, \quad I(z_i; y \mid h_i), \quad (18)$$

where $I(z_i; y \mid h_i)$ denotes conditional mutual information when such a quantity is estimated. These quantities can characterize local activity or utilization; they do not by themselves prove a particular uncertainty semantics or general predictive advantage.

4.6 When is a variational neuron Bayesian?

Three levels clarify the terminology.

Level 1: variationally regularized stochastic architecture. Weights are point-estimated, each unit uses $q_{\phi_i}(z_i \mid h_i)$ and a KL penalty to a reference, and prediction may use samples or means. This is not automatically a standard BNN.

Level 2: Bayesian inference over local latent variables. A coherent joint density may be written as

$$p_{\Theta}(y, z_{1:L} \mid x) = p_{\Theta}(y \mid h_L, z_L) \prod_{\ell=1}^L p_{\psi_{\ell}}(z_{\ell} \mid h_{\ell}, c_{\ell}), \quad (19)$$

with an inference family such as $q_{\Phi}(z_{1:L} \mid x, y)$ and prediction integrating over local states. Parameters Θ may remain point-estimated.

Level 3: hierarchical Bayesian EVE network. Priors are also placed over weights or functions:

$$w \sim p(w), \quad (20)$$

$$z_i \sim p_{\psi_i}(z_i \mid h_i, w, c_i), \quad (21)$$

$$y \sim p(y \mid x, w, \mathbf{z}). \quad (22)$$

Prediction marginalizes both global and local unknowns:

$$p(y_* \mid x_*, \mathcal{D}) = \iint p(y_* \mid x_*, w, z_*) p(z_* \mid x_*, w) p(w \mid \mathcal{D}) dz_* dw. \quad (23)$$

This construction is simultaneously Bayesian, variational when VI is used, stochastic, and distributional at the unit level. The properties remain conceptually distinct.

5 Comparison with Related Probabilistic Architectures

EVE is characterized by the joint specification of a local random object, conditioning, reference distribution, decomposable cost, forward role, and indexed diagnostics. Relative to earlier work [41], this article formalizes that unit-level criterion.

5.1 Latent-variable models, stochastic units, and bottlenecks

VAEs and stochastic backpropagation established amortized inference for input-conditioned latent variables [27,39]. Deep exponential families, variational recurrent neural networks, and ladder VAEs distribute latent variables across depth or time [13,36,44]. Their latents are usually organized by example, layer, hierarchy, or sequence. A variational neuron uses analogous machinery at a declared unit or small block.

Stochastic hidden units predate modern VI and include randomized activations, gates, and conditional-computation variables [7,23,54,30,45,47]. Sampling a hidden variable is not sufficient for the variational-neuron label: the local distribution, reference, cost, conditioning, and forward role must also be identified.

The deep variational information bottleneck, Information Dropout, and the Markov Information Bottleneck provide especially close antecedents through input-conditioned stochastic representations and decomposable KL-like constraints [2,3,58]. Probabilistic embeddings similarly represent inputs by distributions [34,43]. These methods become unit-level equivalents only when the declared granularity and local quantities satisfy Definition 1; otherwise they remain representation-, layer-, or embedding-level constructions.

5.2 Weight uncertainty, dropout, and local reparameterization

Variational BNNs place approximate posteriors over weights [11,18]. The local reparameterization trick may sample pre-activations, but the inferential random object is still the weight posterior [26]. Dropout can be interpreted as regularization, model averaging, or, under particular assumptions, approximate variational Bayesian inference [16]. In all cases, the object specified by the joint model takes precedence over the location at which equivalent noise is sampled.

5.3 Output distributions and final-layer inference

Heteroscedastic regression predicts parameters of an observation distribution and is distributional at the predictive interface rather than inside the hidden unit [25]. Variational Inference on the Final-Layer Output (VIFO) and Variational Classification place variational inference at the final-layer or pre-softmax interface [66,62]. These constructions are close comparators because they introduce probabilistic structure near prediction while leaving the hidden-unit locus distinct.

5.4 Distribution-valued hidden representations and Transformers

Assumed-density, expectation-propagation, moment-propagation, and probabilistic backpropagation methods transform approximate beliefs through neural computations [49,52,19,40,60,61]. Analytic covariance propagation, Gaussian-mixture propagation, output-distribution bounds, and probabilistic state-space layers extend this family [51,55,56,57,63]. Their defining question is

what distributional representation crosses layers. By contrast, a local variational neuron may produce an ordinary point-valued activation after using its latent state; the propagated object must be reported rather than inferred from the presence of a distribution inside the unit.

Probabilistic Transformers also occupy several loci. Bayesian Transformer language models and BayesFormer place uncertainty over parameters or use dropout-related approximations [59,64]; Della introduces layer-wise latent variables [53]; variational attention randomizes an attention representation [65]; and Gaussian-process Transformer approaches attach probabilistic semantics to attention kernels or units [50]. None of these labels alone identifies a unit-level variational neuron; the six reporting questions remain necessary.

Additional neighboring architectures are summarized in Appendix A (Table 4).

6 Controlled Empirical Illustration

To isolate architectural locus, we compare four models: a scalar deterministic baseline (DET-scalar), an output-heteroscedastic stochastic comparator (STOCH-hetero), a capacity-matched deterministic cell (DET-cell matched), and EVE. The benchmark is designed to isolate the effect of architectural locus under controlled, capacity-matched conditions.

6.1 Protocol and compared architectures

Complete data-generating equations, split construction, optimization settings, Monte Carlo sample counts, and model specifications are reported in Appendix B. Here we retain only the design features required to interpret the architectural control. The one-dimensional heteroscedastic regression benchmark is evaluated over five independently generated train/validation/test splits.

DET-scalar is a deterministic point predictor with a fitted global Gaussian observation scale. STOCH-hetero predicts an input-dependent output mean and log-variance but contains no local latent state or local KL term. DET-cell matched is a deterministic encoder-decoder hidden cell with 2,536 parameters. EVE places a diagonal-Gaussian latent of dimension $k = 2$ inside the hidden cell and has 2,472 parameters; the deterministic control therefore has 64 more parameters. Under the distinction introduced in Section 4.4, the experimental EVE instance is an EVE-reg unit: it uses local variational regularization and does not claim strict local posterior inference.

6.2 Results

On this controlled benchmark, EVE and DET-cell matched attain the same mean MSE (0.0776) under approximately matched capacity; EVE has 2,472 trainable parameters versus 2,536 for the deterministic control. Their probabilistic behavior nevertheless differs substantially: EVE attains a mean NLL of -0.3014

Table 3. Five-seed test results for point accuracy and probabilistic scores. Values are mean $\pm$ between-seed standard deviation; lower is better. Coverage, interval width, and all descriptive intervals are reported in Appendix B.

Model	MSE $\downarrow$	NLL $\downarrow$	CRPS $\downarrow$
DET-scalar	0.0793 $\pm$ 0.0064	0.1593 $\pm$ 0.0489	0.1448 $\pm$ 0.0045
STOCH-hetero	0.0816 $\pm$ 0.0062	-0.2748 $\pm$ 0.0341	0.1303 $\pm$ 0.0043
DET-cell matched	0.0776 $\pm$ 0.0064	0.1461 $\pm$ 0.0459	0.1425 $\pm$ 0.0044
EVE $k = 2$	0.0776 $\pm$ 0.0061	-0.3014 $\pm$ 0.0326	0.1276 $\pm$ 0.0040

versus 0.1461 and a mean CRPS of 0.1276 versus 0.1425. Thus, at approximately matched capacity, replacing a deterministic hidden cell by an internal distributional latent state changes the model’s probabilistic behavior while preserving point accuracy. This benchmark-specific contrast therefore isolates the relevance of computational locus. EVE’s narrower interval and slightly lower coverage characterize a sharpness–coverage trade-off compatible with mild under-dispersion.

STOCH-hetero provides a strong output-level comparator, whereas EVE places its distribution inside hidden computation. Similar probabilistic scores can therefore arise from different computational loci.

The unit-level diagnostics further show that addressability is operational rather than merely terminological. Across the five seeds, EVE’s local latent variance tracks the known heteroscedastic structure, with $\text{corr}(\sigma_i^2(x), \sigma_{\text{true}}^2(x)) = 0.941 \pm 0.035$. In a separate training-time local-control grid at fixed latent dimensionality $k = 2$, changing the variational coefficient from $\beta = 5 \times 10^{-2}$ to $\beta = 10^{-4}$ changes mean local KL from 0.056 to 4.280 and mean latent energy $\|\mu_i\|^2$ from 0.029 to 7.909, while test MSE (0.0776 versus 0.0779), CRPS (0.1275 versus 0.1278), and mean predictive standard deviation (0.2252 versus 0.2275) remain similar. These separately trained configurations demonstrate training-time control over empirically distinguishable internal variational regimes while maintaining similar output-level behavior.

7 Discussion and Terminological Recommendations

7.1 What does a local latent distribution represent?

Before explicit identification, $\sigma_i^2(h_i)$ and $\text{KL}_i(h_i)$ should be described as *internal latent dispersion*, *local variational activity*, or *representation uncertainty*. A local state may react to observation noise, ambiguity, distribution shift, model conflict, capacity limits, or optimization pressure. The words *epistemic* and *aleatoric* require additional modelling assumptions or empirical identification.

A hybrid model can separate variation over weights from variation over local states:

$$\tilde{p}_q(y \mid x, \mathcal{D}) := \iint p(y \mid x, w, z) q_\lambda(w \mid \mathcal{D}) q_\phi(z \mid x, w) dz dw. \quad (24)$$

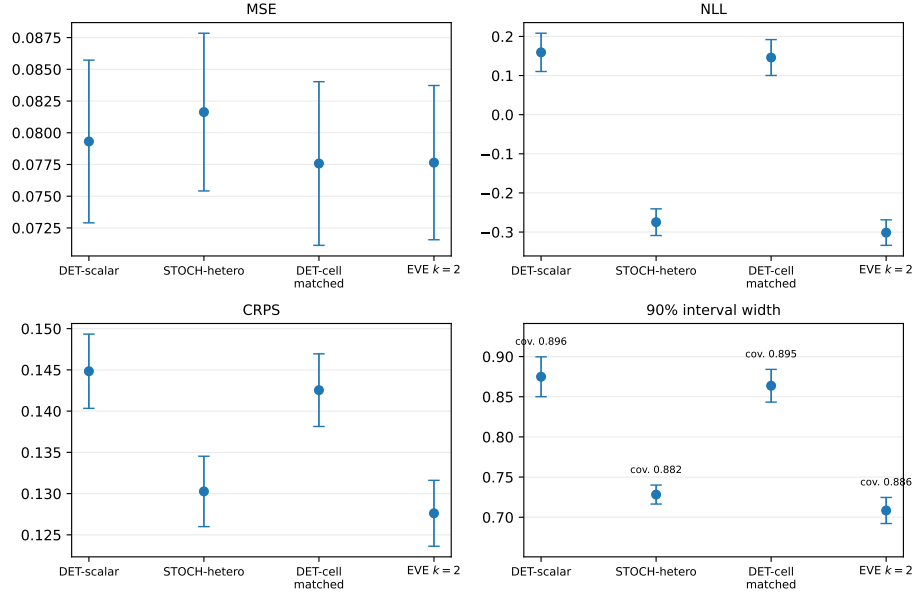


Fig. 2. Five-seed capacity-matched heteroscedastic benchmark. Lower is better for MSE, NLL, and CRPS. Interval width must be interpreted jointly with empirical coverage. Coverage values for nominal 90% intervals are DET-scalar 0.896, STOCH-hetero 0.882, DET-cell matched 0.895, and EVE $k=2$ 0.886.

Variation across w samples concerns the approximate posterior over models; variation across z samples concerns the local state conditional on the input and model.

An aleatoric interpretation is available when the latent participates in an observation model such as

$$z \sim p_\psi(z \mid x), \quad y \sim p_\theta(y \mid x, z), \quad (25)$$

and the resulting conditional variability is identified relative to the likelihood. Calibration remains separate from semantics and should be evaluated through proper scoring rules, coverage, risk–coverage analysis, and robustness under shift [17,35].

7.2 Recommended names

The following labels expose the probabilistic commitments of a construction:

1. *Bayesian neural network*: a model with posterior uncertainty over weights or functions and posterior-predictive marginalization, unless a broader Bayesian latent-variable meaning is stated explicitly.

2. *VI-BNN*: variational inference over weight or function uncertainty.
3. *Variational latent neuron*: a local amortized distribution approximating a posterior under an explicit latent-variable model.
4. *Variationally regularized distributional neuron* or *EVE-reg*: an input-conditioned local distribution regularized relative to a reference, with point-estimated weights and no automatic posterior claim.
5. *Stochastic hidden unit*: a randomized hidden computation whose prior-posterior semantics, if any, are stated separately.
6. *Heteroscedastic output model*: an architecture predicting input-dependent observation-distribution parameters.
7. *Bayesian EVE*: an explicit Bayesian latent-variable formulation over local states.
8. *BNN+EVE*: simultaneous model-level and local-state uncertainty.
9. *Internal latent variability*: the default uncertainty term until epistemic or aleatoric semantics are established.

7.3 Scope

This study establishes unit-level probabilistic computation through explicit membership and granularity criteria and evaluates it in a controlled five-seed benchmark. The capacity-matched design isolates the architectural locus and demonstrates an operationally distinct and measurable computational identity for EVE. This framework provides a basis for extending the analysis across additional tasks, architectures, and data regimes.

8 Conclusion

Bayesian neural networks and variational neurons may share priors, KL terms, reparameterization, and Monte Carlo computation while differing in random object and locus. BNNs marginalize dataset-conditioned model uncertainty; EVE uses input-conditioned local latents inside declared hidden units. The four-axis framework and addressability criteria make this distinction operational. On the controlled benchmark, EVE and the capacity-matched deterministic cell attain the same mean MSE while exhibiting markedly different probabilistic scores, illustrating that the hidden-unit distributional locus can change probabilistic behavior without a corresponding change in point accuracy.

The framework therefore makes explicit what is random, where it is inferred or regularized, and how it enters neural computation.

Data and Code Availability The data-generating equation, split construction, seeds, optimization settings, aggregate results, and descriptive uncertainty intervals are specified in Section 6 and Appendix B. The code and generated outputs for the controlled benchmark are available at <https://github.com/jwayu-com/eve-variational-neurons-reproducibility>. An archival version of the article is available at <https://doi.org/10.5281/zenodo.21870089>.

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A Extended Positioning of Related Architectures

Table 4. Complete positioning relative to neighboring probabilistic architectures. The decisive comparison is the modeled random object, its conditioning, and its locus.

Family	Primary object and conditioning	Principal role	Distinction from a unit-level variational neuron
Standard BNN	Weights or functions conditioned on dataset $\mathcal{D}$	Posterior model averaging	Global model-level inferential object rather than a per-input local hidden state
Deep or sequential latent model	Global, layered, or temporal latents	Generative modelling or stochastic dynamics	Usually organized at example, layer, hierarchy, or sequence granularity
Stochastic hidden-unit network	Hidden variable or gate sampled during computation	Multimodal prediction or conditional computation	Sampling alone does not establish local variational semantics or decomposable reference cost
Variational bottleneck or probabilistic embedding	Input-conditioned representation distribution	Compression, invariance, or metric learning	Typically representation- or embedding-level unless unit granularity is explicitly declared
Variational BNN with local reparameterization	Weight posterior represented through sampled pre-activations	Efficient gradient estimation	Local sample, but global weight-space random object
Probabilistic state-propagation network	Distributional input, activation, or approximate belief	Layerwise transformation of uncertainty	Focuses on propagated distributional state; local latent identity and cost may differ
Final-layer variational method	Output or pre-softmax latent	Output-space inference and calibration	Predictive-interface locus rather than hidden-unit locus
Heteroscedastic, mixture, prior-network, or evidential output [8,32,4,42]	Observation distribution or output-parameter distribution	Predictive density or output uncertainty	Distribution appears at the output interface
Ensemble or mixture of experts [28,22]	Whole-model member or routing variable	Model diversity or conditional routing	Whole-model or routing locus; compatible but non-equivalent
Distributional reinforcement learning [6]	Return distribution	Distributional value or return prediction	Task-output locus rather than a local hidden-state inferential object
Variational neuron	Local latent z_i conditioned on h_i, c_i	Local distributional hidden computation	Declared unit, local reference/cost, forward dependence, and indexed diagnostics

B Benchmark Specification and Evaluation Protocol

B.1 Data-generating process and protocol

For each split,

$$x \sim \mathcal{U}[-4, 4], \quad (26)$$

$$y \mid x \sim \mathcal{N}(m(x), \sigma^2(x)), \quad (27)$$

$$m(x) = 0.85 \tanh(1.35x) + 0.10x, \quad (28)$$

$$\sigma(x) = 0.07 + 0.52 \exp\left[-\frac{1}{2} \left(\frac{x}{0.85}\right)^2\right] + 0.05 (1 + \exp[-3(|x| - 2.7)])^{-1}. \quad (29)$$

The conditional mean is smooth, with a central high-variance region and mild edge variance.

The experiment uses model seeds $s \in \{0, 1, 2, 3, 4\}$. For each seed, independent training, validation, and test sets contain 2,500, 900, and 1,500 observations. Generator seeds are $10000 + 10s + 1$, $10000 + 10s + 2$, and $10000 + 10s + 3$. Models are trained on a CUDA-enabled GPU for 1,200 full-batch epochs with Adam, learning rate 2×10^{-3} , and gradient clipping at Euclidean norm 5. Validation loss is checked every 25 epochs, including the first and last epoch, and the minimum-validation-loss checkpoint is retained. Stochastic and variational models use 32 Monte Carlo samples during training and 160 during evaluation.

DET-scalar is a scalar deterministic predictor with one fitted global Gaussian observation scale. STOCH-hetero predicts an input-dependent target mean and log-variance but contains no local latent variable, prior, or local KL term. DET-cell matched uses a deterministic encoder-decoder activation cell with 2,536 trainable parameters. EVE uses a diagonal-Gaussian local latent state with $k = 2$, hidden width 32, $p(z) = \mathcal{N}(0, I)$, $\beta = 10^{-3}$, soft-band coefficient 0.05, and activity band $[0.01, 2.0]$; it contains 2,472 trainable parameters. The deterministic cell therefore has 64 more parameters than EVE.

Model architectures. All hidden nonlinearities are tanh. STOCH-hetero uses a 1-32-32-1 network for its input-dependent variance. DET-cell matched uses a 1-32-32-4 encoder and a 5-32-32-1 decoder. For EVE with $k = 2$, a 1-32-32-4 encoder parameterizes $(\mu, \log \sigma^2)$, and a 3-32-32-1 decoder maps $[z, h]$ to the activation.

Predictive evaluation. All reported predictive metrics are computed using $S = 160$ predictive samples per input, including observation noise. DET-scalar and DET-cell matched use a learned global Gaussian observation variance, STOCH-hetero uses its learned input-dependent Gaussian variance, and EVE samples its local latent state before decoding and adding learned global Gaussian observation noise. MSE is computed from the predictive sample mean. NLL denotes the Gaussian negative log-likelihood of the moment-matched predictive mean and variance estimated from these samples. CRPS is computed directly from the predictive samples, and nominal 90% intervals are their empirical 5th and 95th percentiles.

Unit-level diagnostics and training-time local control. For the main EVE model, local latent variance, local KL, and latent mean energy are evaluated on the input grid for each seed. The reported correlation between local latent variance and the known conditional standard deviation is 0.941 ± 0.035 across seeds. A separate control grid uses the same five seeds and 500 training epochs per configuration. The fixed- $k = 2$, hidden-width-32 comparison used in Section 6 contrasts $\beta = 5 \times 10^{-2}$ and $\beta = 10^{-4}$ with no soft-band penalty in either configuration. Their respective mean values are: local KL 0.056 and 4.280, $\|\mu_i\|^2$ 0.029 and 7.909, MSE 0.0776 and 0.0779, CRPS 0.1275 and 0.1278, and mean predictive standard deviation 0.2252 and 0.2275. Because each configuration is retrained, these results establish training-time local control and observability, not a same-checkpoint causal intervention.

B.2 Complete numerical results

Table 5. Complete five-seed test results. Values are mean $\pm$ between-seed standard deviation. Lower is better for MSE, NLL, and CRPS; width is interpreted jointly with coverage.

Model	MSE $\downarrow$	NLL $\downarrow$	CRPS $\downarrow$
DET-scalar	0.0793 ± 0.0064	0.1593 ± 0.0489	0.1448 ± 0.0045
STOCH-hetero	0.0816 ± 0.0062	-0.2748 ± 0.0341	0.1303 ± 0.0043
DET-cell matched	0.0776 ± 0.0064	0.1461 ± 0.0459	0.1425 ± 0.0044
EVE $k = 2$	0.0776 ± 0.0061	-0.3014 ± 0.0326	0.1276 ± 0.0040

Model	Coverage 90%	Width 90%
DET-scalar	0.8963 ± 0.0099	0.8749 ± 0.0248
STOCH-hetero	0.8819 ± 0.0090	0.7282 ± 0.0118
DET-cell matched	0.8951 ± 0.0102	0.8637 ± 0.0204
EVE $k = 2$	0.8856 ± 0.0082	0.7084 ± 0.0162

For descriptive uncertainty over the five seed-level means, two-sided 95% Student- t intervals are: for DET-cell matched, MSE [0.0696, 0.0856], NLL [0.0891, 0.2030], CRPS [0.1371, 0.1480], coverage [0.8825, 0.9077], and width [0.8384, 0.8890]; for EVE, MSE [0.0701, 0.0852], NLL $[-0.3419, -0.2609]$, CRPS [0.1227, 0.1326], coverage [0.8754, 0.8958], and width [0.6883, 0.7286]. These intervals are descriptive summaries of variability across the five runs and are not used for hypothesis testing.

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